POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 2032 | Semester II | Theory | 4 Credits

Electric Circuits and Systems

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electrical & Electronics Engineering, Electrical Engineering and Electrical Engineering & Electric Vehicles Technology**
Contact hours	**60**
Teaching scheme	**4:0:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**6 h/week; 90 h total**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

This handbook is based only on the supplied official **Diploma Curriculum Revision 2026** syllabus PDF for Course **2032**, titled **Electric Circuits and Systems**. Objectives, course outcomes, teaching scheme, major topics, detailed coverage, activities and assessment are drawn from that source.

Source treatment: PDF table wrapping can interrupt a heading across lines. In such cases, this handbook retains the official intent in a clearly labelled topic. Normal teaching explanations, study flows, diagrams and Malayalam-support boxes are handbook support rather than claims of official wording.

Verified source note: No web research was used to establish syllabus content. Official references and URLs appear only where they are listed in the supplied PDF.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Electric Circuits and Systems develops calculation and representation skills for DC network laws/theorems, AC series and parallel circuits, resonance and balanced three-phase systems. It requires clear phasor, waveform and circuit reasoning.

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Prerequisite foundation

Fundamentals of Engineering Mathematics (1002) and Basic Electrical & Electronics Engineering (1031).

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
4	0	0	0	6 h/week; 90 h total	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Use basic laws and theorems to solve circuits.
2. Analyse AC series and parallel circuits.
3. Calculate resonance and parallel-AC parameters.
4. Analyse three-phase systems and interconnections.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

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Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Solve electric circuits using basic laws and network theorems.	Apply
CO2	Apply AC-series-circuit knowledge for different loads.	Apply
CO3	Apply resonance and methods for AC parallel circuits.	Apply
CO4	Calculate three-phase balanced-system parameters with different interconnections.	Apply

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Electric Circuits and Network Theorems	Circuit types/terms, Kirchhoff laws, mesh/nodal analysis, network theorems and star-delta conversion.	3.5 L + 11.5 T hours	CO1	Module 1
2	Phasor Algebra and AC Series Circuits	j operator, phasor rectangular/polar conversion and operations; pure R/L/C, RL/RC/RLC, impedance, power and power factor.	10 L + 5 T hours	CO2	Module 2
3	Resonance and Parallel AC Circuits	Series resonance, Q factor and bandwidth; admittance/conductance/susceptance; two-branch parallel circuits; parallel resonance and applications.	7 L + 8 T hours	CO3	Module 3
4	Three-Phase AC Systems	Three-phase generation/waveforms/phasors/sequence; star/delta connections; balanced-system calculations and power.	9.5 L + 5.5 T hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Electric Circuits and Network Theorems

Circuit types/terms, Kirchhoff laws, mesh/nodal analysis, network theorems and star-delta conversion.

CO1 · 3.5 L + 11.5 T hours

Phasor Algebra and AC Series Circuits

j operator, phasor rectangular/polar conversion and operations; pure R/L/C, RL/RC/RLC, impedance, power and power factor.

CO2 · 10 L + 5 T hours

Resonance and Parallel AC Circuits

Series resonance, Q factor and bandwidth; admittance/conductance/susceptance; two-branch parallel circuits; parallel resonance and applications.

CO3 · 7 L + 8 T hours

Three-Phase AC Systems

Three-phase generation/waveforms/phasors/sequence; star/delta connections; balanced-system calculations and power.

CO4 · 9.5 L + 5.5 T hours

MODULE 1

Electric Circuits and Network Theorems

Circuit types/terms, Kirchhoff laws, mesh/nodal analysis, network theorems and star-delta conversion.

3.5 L + 11.5 T hours **CO1** **Understand · Apply**

Learning goals

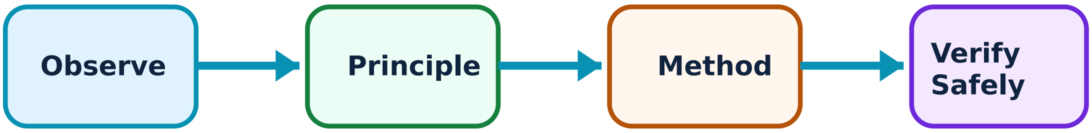
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Electric Circuits and Network Theorems: identify the system, state its principle, follow the correct method and verify the result safely.

1. Circuit fundamentals and Kirchhoff laws−

Circuit analysis uses branch, node, loop, current, voltage, source and reference direction consistently. Kirchhoff’s current law expresses charge conservation at a node; Kirchhoff’s voltage law expresses algebraic voltage balance around a loop.

Study and application method

Assign current/voltage directions, write equations with a consistent sign convention, solve maximum-three-loop systems and interpret negative result as reversed assumed direction.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Assign current/voltage directions, write equations with a consistent sign convention, solve maximum-three-loop systems and interpret negative result as reversed assumed direction.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not mix reference directions midway through an equation set.

2. Mesh and nodal analysis−

Mesh analysis uses loop currents; nodal analysis uses node voltages. Both transform a circuit into simultaneous equations and are limited to the stated complexity in the syllabus.

Study and application method

Choose method that gives fewer unknowns, label circuit clearly, write independent equations and verify one result using KCL/KVL.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Choose method that gives fewer unknowns, label circuit clearly, write independent equations and verify one result using KCL/KVL.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: A current source or voltage source may require special treatment; identify it before writing equations.

3. Network theorems and star-delta conversion−

Superposition, Thevenin, Norton and maximum-power-transfer theorems simplify linear-network analysis. Star-delta conversion replaces a three-terminal network with an equivalent form for simple DC circuits.

Study and application method

State theorem, deactivate only independent sources where appropriate, determine equivalent seen at load terminals and then restore load/result.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: State theorem, deactivate only independent sources where appropriate, determine equivalent seen at load terminals and then restore load/result.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Superposition applies to voltage/current, not directly to power; calculate power after final response.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Phasor Algebra and AC Series Circuits

j operator, phasor rectangular/polar conversion and operations; pure R/L/C, RL/RC/RLC, impedance, power and power factor.

10 L + 5 T hours **CO2** **Understand · Apply**

Learning goals

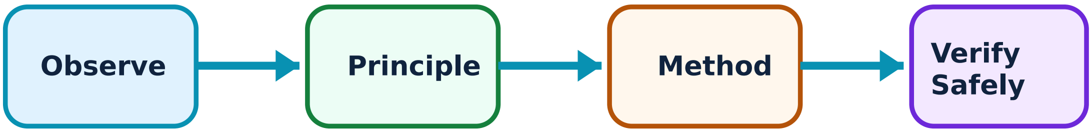
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Phasor Algebra and AC Series Circuits: identify the system, state its principle, follow the correct method and verify the result safely.

1. Phasors and complex-number representation−

A phasor represents sinusoidal magnitude and phase at common frequency. The j operator represents a ninety-degree rotation. Rectangular and polar forms support different calculations.

Study and application method

Convert consistently between forms, add/subtract in rectangular form and multiply/divide in polar form, then state final magnitude/angle.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Convert consistently between forms, add/subtract in rectangular form and multiply/divide in polar form, then state final magnitude/angle.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not add magnitudes directly when phase angles differ.

2. Pure R, L and C AC circuits−

In pure resistance, voltage/current are in phase. In inductance, current lags voltage; in capacitance, current leads voltage. Phasor diagrams and waveforms express these relationships.

Study and application method

Draw one reference phasor, mark lead/lag by 90°, calculate reactance/impedance and identify power implication.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw one reference phasor, mark lead/lag by 90°, calculate reactance/impedance and identify power implication.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Reactance is frequency dependent; do not treat it as fixed DC resistance.

3. RL, RC and RLC series circuits and power−

Series AC circuits combine resistance and reactance vectorially. Impedance triangle, power triangle and power factor relate voltage/current phase difference to active, reactive and apparent power.

Study and application method

Compute R and net reactance, form impedance, find current and then calculate P, Q, S and power factor with units.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compute R and net reactance, form impedance, find current and then calculate P, Q, S and power factor with units.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Power factor sign/convention must agree with lagging inductive or leading capacitive load.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Resonance and Parallel AC Circuits

Series resonance, Q factor and bandwidth; admittance/conductance/susceptance; two-branch parallel circuits; parallel resonance and applications.

7 L + 8 T hours | CO3 | Understand · Apply

Learning goals

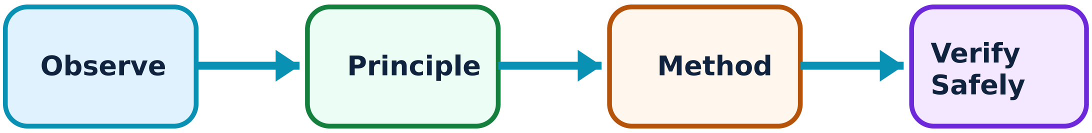
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Resonance and Parallel AC Circuits: identify the system, state its principle, follow the correct method and verify the result safely.

1. Series RLC resonance−

At series resonance inductive and capacitive reactances cancel, leaving minimum impedance in ideal series circuit. Resonant frequency, Q factor and bandwidth describe frequency response.

Study and application method

State resonance condition, calculate resonant frequency/Q from permitted formulas and relate bandwidth to resonant frequency and Q.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: State resonance condition, calculate resonant frequency/Q from permitted formulas and relate bandwidth to resonant frequency and Q.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Resonance does not mean every voltage in a circuit is small; reactive element voltages can be significant.

2. Admittance and parallel-circuit solution−

Admittance is reciprocal of impedance; conductance and susceptance are its real and imaginary components. Parallel circuits are efficiently solved by adding branch admittances or using vector methods.

Study and application method

Convert each branch to admittance, add components, find total current/phase and verify branch-current sum.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Convert each branch to admittance, add components, find total current/phase and verify branch-current sum.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Keep conductance and susceptance signs consistent; capacitive and inductive susceptances oppose.

3. Parallel resonance and comparison−

Parallel resonance has a different impedance/current behaviour from series resonance. The syllabus compares them and asks applications of resonant circuits.

Study and application method

Use a comparison table for impedance, supply current, current/voltage magnification, frequency response and common use.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Use a comparison table for impedance, supply current, current/voltage magnification, frequency response and common use.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not assume a series-resonance graph applies to a parallel circuit.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Three-Phase AC Systems

Three-phase generation/waveforms/phasors/sequence; star/delta connections; balanced-system calculations and power.

9.5 L + 5.5 T hours | CO4 | Understand · Apply

Learning goals

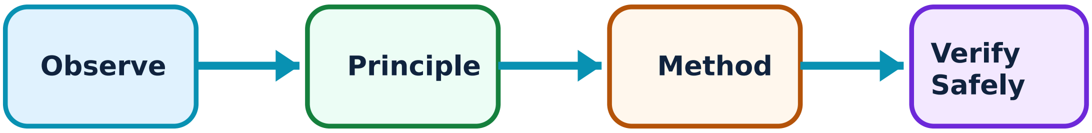
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Three-Phase AC Systems: identify the system, state its principle, follow the correct method and verify the result safely.

1. Three-phase generation and phase sequence−

A three-phase generator produces three equal-frequency voltages displaced by 120°. Waveforms and phasors show sequence and balanced-system relationship. Three-phase systems offer practical advantages in power transmission and machines.

Study and application method

Draw three phase waveforms/phasors with sequence marked, then state one operational advantage over single phase.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw three phase waveforms/phasors with sequence marked, then state one operational advantage over single phase.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Changing phase sequence can reverse motor rotation; sequence is not only a drawing convention.

2. Star and delta interconnections−

Star and delta connect three phase elements differently. In balanced systems, line/phase voltage and current relations are derived using vector diagrams, and networks are solved from these relations.

Study and application method

State connection, identify line/phase quantities, draw vector relation and use correct √3 factor with a labelled derivation.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: State connection, identify line/phase quantities, draw vector relation and use correct √3 factor with a labelled derivation.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not interchange star voltage relation with delta current relation.

3. Three-phase power−

Balanced three-phase systems have active, reactive and apparent power related to line quantities and power factor. The same physical definitions apply but with three-phase relation.

Study and application method

Identify balanced connection, calculate requested line/phase value, then use appropriate power expression and state unit/power factor.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Identify balanced connection, calculate requested line/phase value, then use appropriate power expression and state unit/power factor.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Use line values only with the matching formula; show whether data are line or phase quantities.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Electric Circuits and Network Theorems

Use the concept flow to revise observation, principle, method and verification.

Module 2: Phasor Algebra and AC Series Circuits

Use the concept flow to revise observation, principle, method and verification.

Module 3: Resonance and Parallel AC Circuits

Use the concept flow to revise observation, principle, method and verification.

Module 4: Three-Phase AC Systems

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Solve KCL/KVL, mesh/nodal, network-theorem and star-delta problems.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Practise polar/rectangular phasor operations and AC power-factor calculations.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Compare series/parallel resonance and solve two-branch circuits by admittance/vector methods.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw three-phase phasors and calculate balanced star/delta system parameters and power.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure		
Component	**Official mark / conversion**	**Student evidence**
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Electric Circuits and Network Theorems

Explain Circuit fundamentals and Kirchhoff laws and state one correct method, application or precaution.−

Circuit analysis uses branch, node, loop, current, voltage, source and reference direction consistently. Kirchhoff’s current law expresses charge conservation at a node; Kirchhoff’s voltage law expresses algebraic voltage balance around a loop.

Answer framework: Assign current/voltage directions, write equations with a consistent sign convention, solve maximum-three-loop systems and interpret negative result as reversed assumed direction.

Important point: Do not mix reference directions midway through an equation set.

Explain Mesh and nodal analysis and state one correct method, application or precaution.−

Mesh analysis uses loop currents; nodal analysis uses node voltages. Both transform a circuit into simultaneous equations and are limited to the stated complexity in the syllabus.

Answer framework: Choose method that gives fewer unknowns, label circuit clearly, write independent equations and verify one result using KCL/KVL.

Important point: A current source or voltage source may require special treatment; identify it before writing equations.

Explain Network theorems and star-delta conversion and state one correct method, application or precaution.−

Superposition, Thevenin, Norton and maximum-power-transfer theorems simplify linear-network analysis. Star-delta conversion replaces a three-terminal network with an equivalent form for simple DC circuits.

Answer framework: State theorem, deactivate only independent sources where appropriate, determine equivalent seen at load terminals and then restore load/result.

Important point: Superposition applies to voltage/current, not directly to power; calculate power after final response.

Module 2 — Phasor Algebra and AC Series Circuits

Explain Phasors and complex-number representation and state one correct method, application or precaution.−

A phasor represents sinusoidal magnitude and phase at common frequency. The j operator represents a ninety-degree rotation. Rectangular and polar forms support different calculations.

Answer framework: Convert consistently between forms, add/subtract in rectangular form and multiply/divide in polar form, then state final magnitude/angle.

Important point: Do not add magnitudes directly when phase angles differ.

Explain Pure R, L and C AC circuits and state one correct method, application or precaution.−

In pure resistance, voltage/current are in phase. In inductance, current lags voltage; in capacitance, current leads voltage. Phasor diagrams and waveforms express these relationships.

Answer framework: Draw one reference phasor, mark lead/lag by 90°, calculate reactance/impedance and identify power implication.

Important point: Reactance is frequency dependent; do not treat it as fixed DC resistance.

Explain RL, RC and RLC series circuits and power and state one correct method, application or precaution.−

Series AC circuits combine resistance and reactance vectorially. Impedance triangle, power triangle and power factor relate voltage/current phase difference to active, reactive and apparent power.

Answer framework: Compute R and net reactance, form impedance, find current and then calculate P, Q, S and power factor with units.

Important point: Power factor sign/convention must agree with lagging inductive or leading capacitive load.

Module 3 — Resonance and Parallel AC Circuits

Explain Series RLC resonance and state one correct method, application or precaution.−

At series resonance inductive and capacitive reactances cancel, leaving minimum impedance in ideal series circuit. Resonant frequency, Q factor and bandwidth describe frequency response.

Answer framework: State resonance condition, calculate resonant frequency/Q from permitted formulas and relate bandwidth to resonant frequency and Q.

Important point: Resonance does not mean every voltage in a circuit is small; reactive element voltages can be significant.

Explain Admittance and parallel-circuit solution and state one correct method, application or precaution.−

Admittance is reciprocal of impedance; conductance and susceptance are its real and imaginary components. Parallel circuits are efficiently solved by adding branch admittances or using vector methods.

Answer framework: Convert each branch to admittance, add components, find total current/phase and verify branch-current sum.

Important point: Keep conductance and susceptance signs consistent; capacitive and inductive susceptances oppose.

Explain Parallel resonance and comparison and state one correct method, application or precaution.−

Parallel resonance has a different impedance/current behaviour from series resonance. The syllabus compares them and asks applications of resonant circuits.

Answer framework: Use a comparison table for impedance, supply current, current/voltage magnification, frequency response and common use.

Important point: Do not assume a series-resonance graph applies to a parallel circuit.

Module 4 — Three-Phase AC Systems

Explain Three-phase generation and phase sequence and state one correct method, application or precaution.−

A three-phase generator produces three equal-frequency voltages displaced by 120°. Waveforms and phasors show sequence and balanced-system relationship. Three-phase systems offer practical advantages in power transmission and machines.

Answer framework: Draw three phase waveforms/phasors with sequence marked, then state one operational advantage over single phase.

Important point: Changing phase sequence can reverse motor rotation; sequence is not only a drawing convention.

Explain Star and delta interconnections and state one correct method, application or precaution.−

Star and delta connect three phase elements differently. In balanced systems, line/phase voltage and current relations are derived using vector diagrams, and networks are solved from these relations.

Answer framework: State connection, identify line/phase quantities, draw vector relation and use correct √3 factor with a labelled derivation.

Important point: Do not interchange star voltage relation with delta current relation.

Explain Three-phase power and state one correct method, application or precaution.−

Balanced three-phase systems have active, reactive and apparent power related to line quantities and power factor. The same physical definitions apply but with three-phase relation.

Answer framework: Identify balanced connection, calculate requested line/phase value, then use appropriate power expression and state unit/power factor.

Important point: Use line values only with the matching formula; show whether data are line or phase quantities.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Electric Circuits and Network Theorems.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Phasor Algebra and AC Series Circuits.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Resonance and Parallel AC Circuits.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Three-Phase AC Systems.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Circuit types/terms, Kirchhoff laws, mesh/nodal analysis, network theorems and star-delta conversion..
2. Develop a complete answer or practical plan for: j operator, phasor rectangular/polar conversion and operations; pure R/L/C, RL/RC/RLC, impedance, power and power factor..
3. Develop a complete answer or practical plan for: Series resonance, Q factor and bandwidth; admittance/conductance/susceptance; two-branch parallel circuits; parallel resonance and applications..
4. Develop a complete answer or practical plan for: Three-phase generation/waveforms/phasors/sequence; star/delta connections; balanced-system calculations and power..

LAST-MILE REVISION

Quick Revision

Module 1: Electric Circuits and Network Theorems

Circuit types/terms, Kirchhoff laws, mesh/nodal analysis, network theorems and star-delta conversion.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Phasor Algebra and AC Series Circuits

j operator, phasor rectangular/polar conversion and operations; pure R/L/C, RL/RC/RLC, impedance, power and power factor.

- Match each topic to CO2.
- Redraw or rehearse one key method.

- Check one common mistake/precaution.

Module 3: Resonance and Parallel AC Circuits

Series resonance, Q factor and bandwidth; admittance/conductance/susceptance; two-branch parallel circuits; parallel resonance and applications.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Three-Phase AC Systems

Three-phase generation/waveforms/phasors/sequence; star/delta connections; balanced-system calculations and power.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Circuit fundamentals and Kirchhoff laws	Circuit analysis uses branch, node, loop, current, voltage, source and reference direction consistently.
Mesh and nodal analysis	Mesh analysis uses loop currents; nodal analysis uses node voltages.
Network theorems and star-delta conversion	Superposition, Thevenin, Norton and maximum-power-transfer theorems simplify linear-network analysis.
Phasors and complex-number representation	A phasor represents sinusoidal magnitude and phase at common frequency.
Pure R, L and C AC circuits	In pure resistance, voltage/current are in phase.
RL, RC and RLC series circuits and power	Series AC circuits combine resistance and reactance vectorially.
Series RLC resonance	At series resonance inductive and capacitive reactances cancel, leaving minimum impedance in ideal series circuit.
Admittance and parallel-circuit solution	Admittance is reciprocal of impedance; conductance and susceptance are its real and imaginary components.
Parallel resonance and comparison	Parallel resonance has a different impedance/current behaviour from series resonance.
Three-phase generation and phase sequence	A three-phase generator produces three equal-frequency voltages displaced by 120°.
Star and delta interconnections	Star and delta connect three phase elements differently.
Three-phase power	Balanced three-phase systems have active, reactive and apparent power related to line quantities and power factor.

LEARNING RESOURCES

References

Officially listed print resources

1. B. L. Theraja and A. K. Theraja, A Text Book of Electrical Technology Vol. I.
2. V. K. Mehta and Rohit Mehta, Principles of Electrical Engineering.
3. B. R. Gupta and Vandana Singhal, Fundamentals of Electrical Network.

Officially listed web resources

- https://www.swayam.gov.in
- https://www.electrical4u.com
- https://www.vlab.co.in

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